

AutoRisk

Predicting High-Risk Vehicles from NHTSA

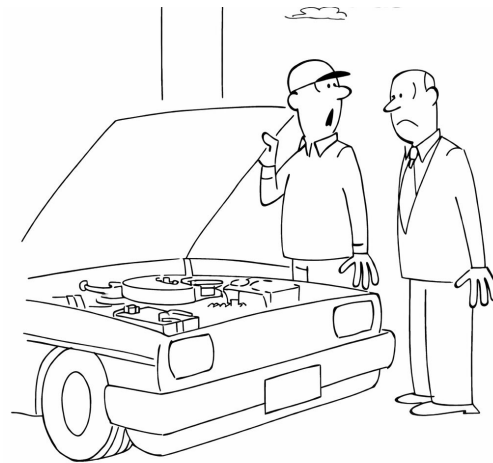
Streamlit App | <https://autorisk-app-nvf6o1mxaa-uc.a.run.app>

Flask Model API | <https://autorisk-api-nvf6o1mxaa-uc.a.run.app>

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05/31/2026

STATS 418 • Spring 2026 • Final Project



Problem Background

Reliability data exists in NHTSA recall + complaint databases, but is fragmented across 50+ endpoints and millions of records.

Goal

Predict whether a (year, make, model) falls in the top 30% of complaint rate, using only metadata + historical recall signals.

Why it matters

Helps buyers, insurers, and dealers screen vehicles. Demonstrates an end-to-end ML pipeline (API → model → API → app) with real, messy public data.

1. Data Sources

NHTSA public APIs :

- vehicle catalog
- Recalls
- complaints



2. Engineering Design

- Local caching to avoid repeated API calls
- Rate limiting and retry mechanism

3. Data Issues & Fixes

- Duplicate trim-level entries detected
- Merged into base models to avoid double-counting

Raw Data

F-150 SuperCab
F-150 Crew Cab
Duplicate complaints

Normalize

→ base_model
SuperCab → F-150
Crew Cab → F-150

Group

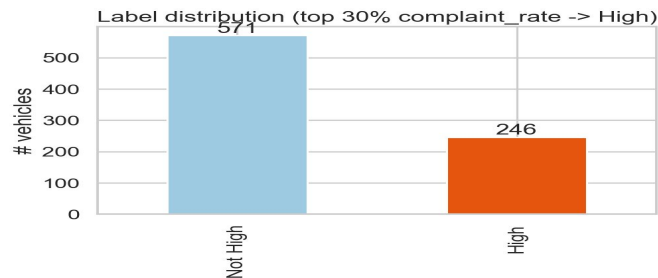
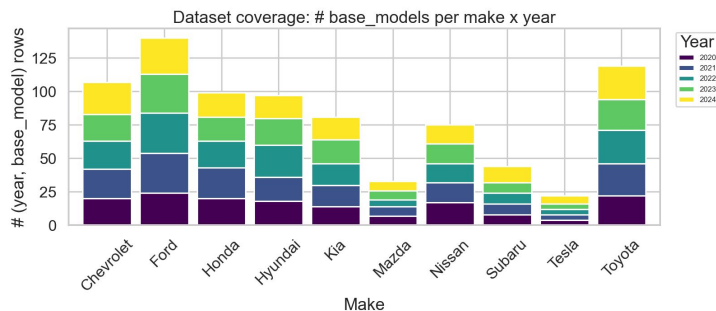
year + make + base_model

Aggregate

Complaint → MAX
Recall → SUM

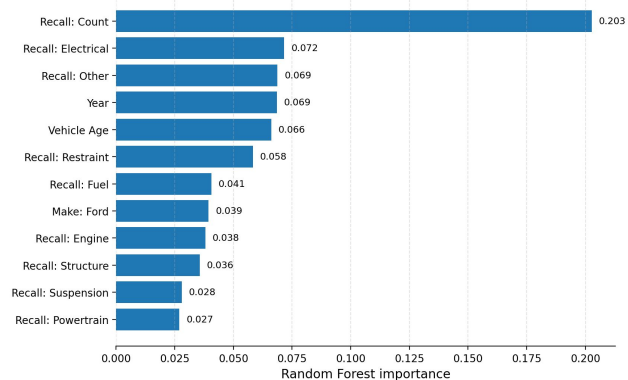
Final Dataset

1 row per vehicle
clean features
complaint_rate

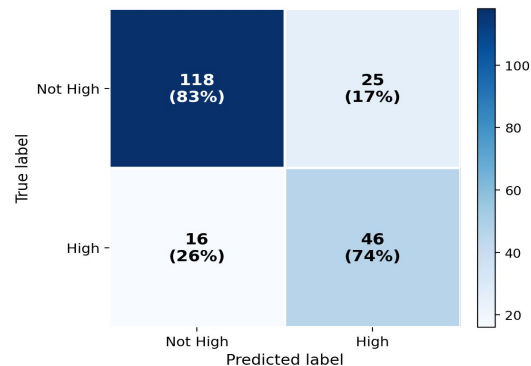


03 — Model Development & Evaluation

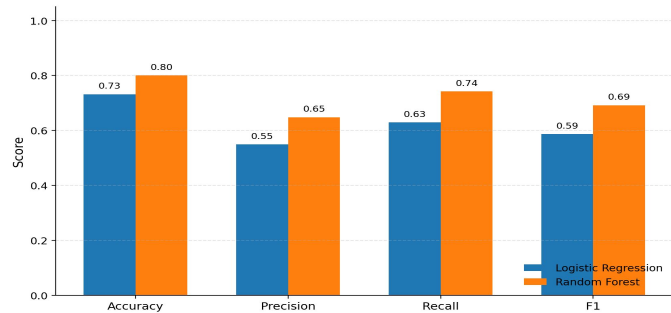
Top Predictive Features



Random Forest Confusion Matrix



Baseline Model Performance



Key Findings

Recall Count was the strongest predictor.

Random Forest consistently outperformed Logistic Regression.

The model correctly identified 74% of high-risk vehicles.

Feature Engineering

Recall counts by category

Vehicle age

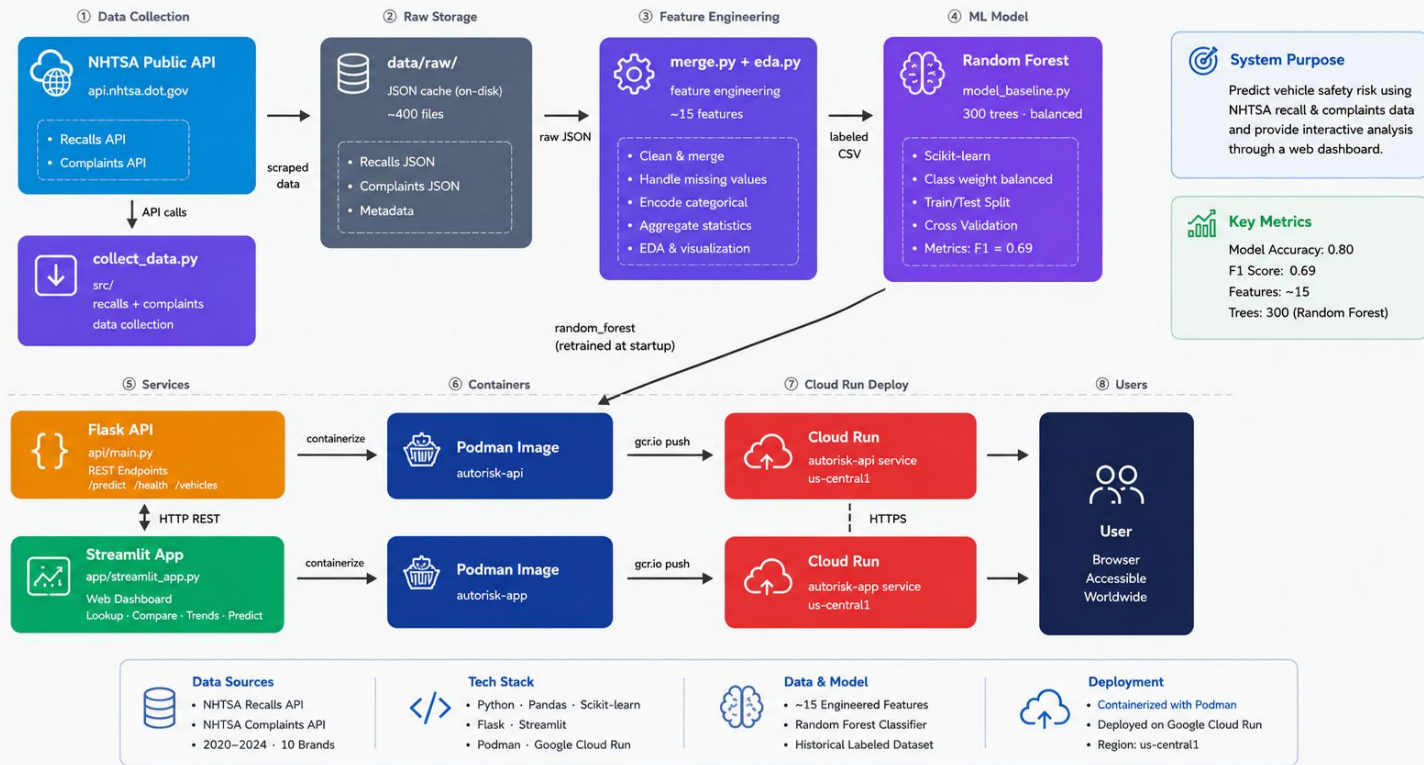
Manufacturer encoding

Complaint statistics

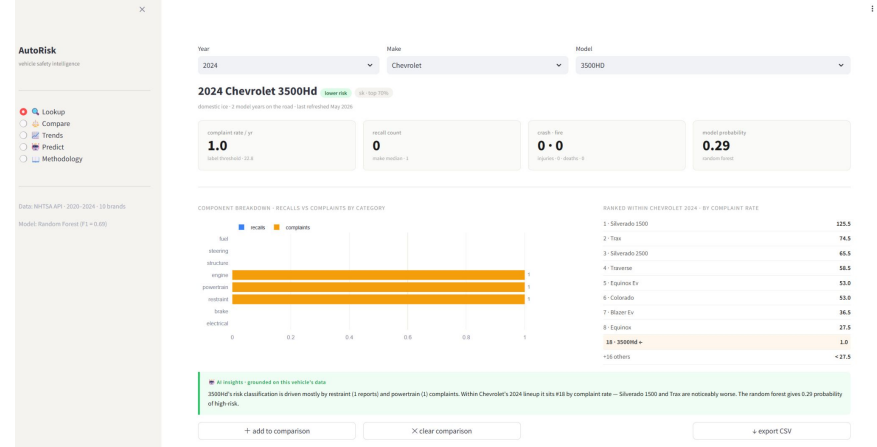
Historical recall signals

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05 — Application Features

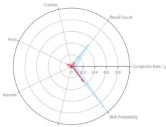


Vehicle Comparison

Rank up to 3 vehicles to compare side by side.

Vehicle 1	Vehicle 2	Vehicle 3
Year	Year	Year
2024	2024	2024
Make	Make	Make
Chevrolet	Ford	Mercedes
Model	Model	Model
3500HD	BRONCO	C300

Vehicle	Complaint Rate / yr	Recall Count	Crashes	Fires	Injuries	Deaths	Risk Probability
2024 Chevrolet 3500HD	1.00	0	0	0	0	0	0.29
2024 Ford Bronco	20.00	20	4	1	1	0	0.48
2024 Mercedes C300	12.00	0	2	0	0	0	0.30



Challenges & Solution :

1. Data Integration
2. Cloud Deployment
3. API Communication

Key learning:

1. Cloud Architecture
2. ML Deployment

Future Development :

1. Ai chat box
2. Add more information
3. Different model (XGBoost)

Role of AI:

1. AI helped me generate a large amount of code, increasing my completion speed by 60%、
2. However, framework-specific issues such as Streamlit widget state management required manual debugging and human intervention to achieve the final solution.